

AUTO LOAN SECURITISATION ANALYTICS USING POWER BI

Executive Summary

This project presents a comprehensive Auto Loan Securitisation Analytics solution developed using Power BI. The dashboard enables monitoring of portfolio exposure, expected losses, delinquency behaviour, vintage performance, IFRS 9 staging, stress testing outcomes, and securitisation waterfall structures.

The solution transforms loan-level data into business intelligence dashboards that support portfolio managers, risk analysts, and investors in evaluating portfolio performance and credit risk. The implementation follows securitisation analytics principles described in the project specification and provides an integrated framework for credit risk monitoring and investor reporting.

Chapter 1: Introduction

Overview of Securitisation

Securitisation is the process of pooling financial assets such as auto loans and converting them into marketable securities. Cash flows generated by the underlying loans are distributed among investors according to predefined payment priorities.

In an auto loan securitisation structure, vehicle loans are pooled together and transferred into a special purpose vehicle (SPV). Investors purchase securities backed by these loan receivables.

Project Objectives

The objectives of this project are:

- Monitor portfolio exposure and balance trends.
 - Track delinquency and default behaviour.
 - Measure expected credit losses.
 - Analyse portfolio performance through vintage analysis.
 - Support IFRS 9 credit risk reporting.
 - Conduct stress testing under adverse economic scenarios.
 - Simulate waterfall cash flow allocation across tranches.
 - Provide investor-focused portfolio reporting.
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Chapter 2: Business Problem

Financial institutions managing securitised portfolios require continuous visibility into asset quality and cash flow performance. Manual reporting methods are often fragmented and inefficient.

The major challenges include:

- Monitoring delinquency migration.
- Estimating expected credit losses.
- Tracking portfolio vintage performance.
- Evaluating stress scenario impacts.
- Allocating cash flows across investor tranches.

This project addresses these challenges through a centralized Power BI analytics platform.

Chapter 3: Dataset Description

Auto Loan Securitisation Dataset

This dataset contains loan-level information used for portfolio analysis.

Key Attributes

- Loan ID
- Borrower ID
- Borrower Age
- Current Balance
- Vehicle Value
- Interest Rate
- Loan Purpose
- Insurance Type
- CIBIL Score
- Delinquency Days

Purpose

Supports portfolio exposure analysis, borrower risk assessment, and credit quality evaluation.

DPD Snapshot History Dataset

This dataset captures monthly delinquency status.

Key Attributes

- Loan ID
- DPD Days
- DPD Bucket
- Amount Overdue
- Cure Flag

Purpose

Supports delinquency analysis and collections monitoring.

Static Pool Vintage Dataset

This dataset supports vintage and cohort analysis.

Key Attributes

- Vintage Month
- Months on Book
- Pool Factor
- Cumulative Net Loss Rate
- Delinquency Metrics

Purpose

Tracks performance of loan cohorts over time.

Dynamic Loss Monthly Dataset

This dataset captures monthly loss metrics.

Key Attributes

- Collection Efficiency
- CPR
- Recoveries
- Beginning Balance
- Ending Balance
- Net Loss

Purpose Supports dynamic loss monitoring and portfolio performance evaluation.

Chapter 4: Data Model Design

Model Architecture

The Power BI model follows a star-schema style design.

Fact Tables

- Auto Loan Securitisation Data
- DPD Snapshot History
- Static Pool Vintage Data
- Dynamic Loss Monthly

Dimension Tables

- Date Table
- Waterfall Steps

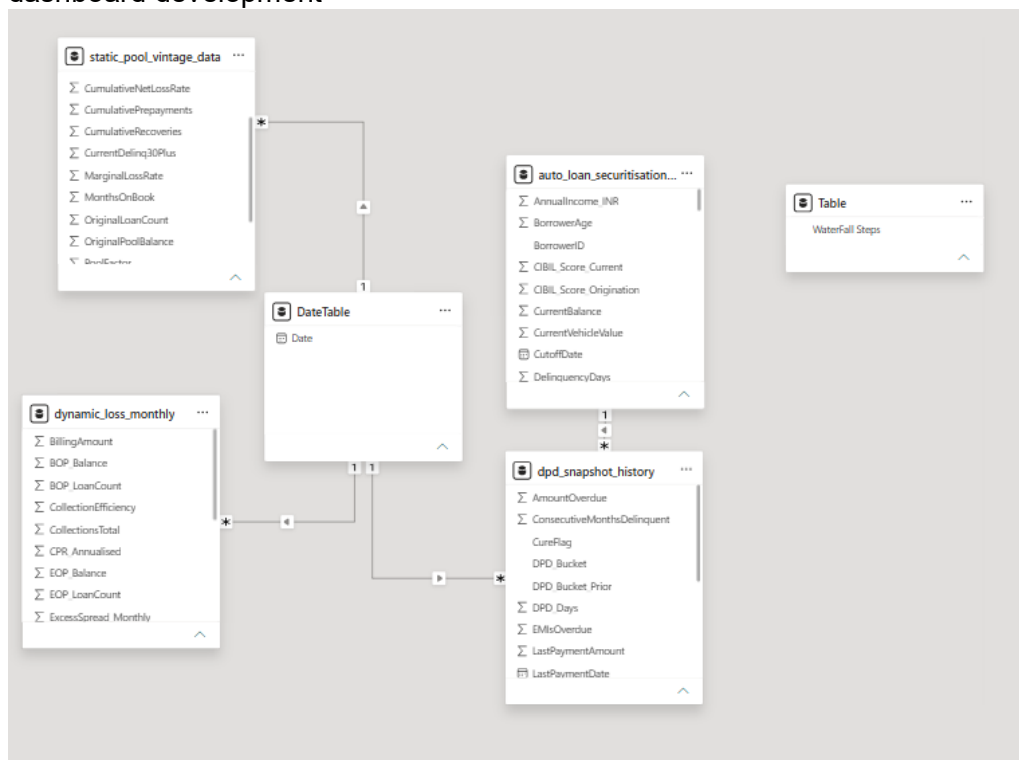
Relationship Design

The Date Table acts as the central time dimension connecting historical delinquency data, vintage data, and monthly loss information. Loan-level relationships support detailed borrower and portfolio analysis.

Model Benefits

- Improved query performance
- Simplified reporting structure Consistent time intelligence calculations Scalable

dashboard development



Chapter 5: DAX Measures and KPI Framework

Total Exposure

Measures total outstanding securitised portfolio balance.

Business Significance: Represents the overall exposure to credit risk.

Expected Loss

Expected Loss estimates anticipated losses from the portfolio based on probability of default assumptions.

Business Significance: Supports risk provisioning and portfolio valuation.

Net Cashflow

Represents expected cash collections after loss adjustments.

Business Significance: Used in waterfall allocation and investor reporting.

Average Probability of Default (PD)

Measures average likelihood of borrower default.

Business Significance: Core credit risk indicator.

Average Loss Given Default (LGD)

Measures percentage loss incurred after default.

Business Significance: Important input for expected loss calculations.

Total Exposure at Default (EAD)

Represents exposure outstanding at the time of borrower default.

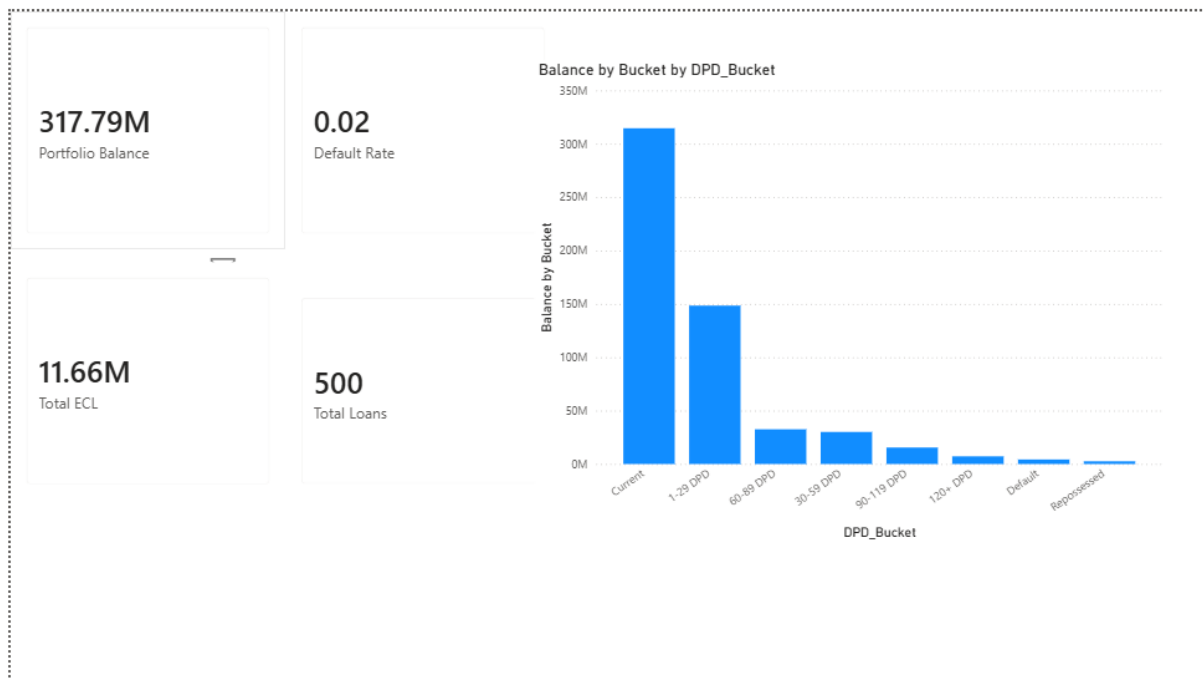
Business Significance: Key IFRS 9 and Basel risk metric.

Total Expected Credit Loss (ECL)

Measures lifetime expected credit losses.

Business Significance: Supports IFRS 9 impairment reporting.

Chapter 6: Executive Dashboard



Objective

Provides a consolidated view of portfolio performance.

Key Metrics

- Total Exposure: 317.79M
- Net Cashflow: 315.50M
- Expected Loss: 2.29M
- Probability of Default

Insights

- Portfolio exposure exceeds 300M.
 - New Vehicle loans represent the majority of balances.
 - Delinquency remains concentrated in lower DPD buckets.
 - Waterfall allocation demonstrates investor payment structure.
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Chapter 7: DPD Analysis Dashboard



Objective

Analyse delinquency behaviour across borrower segments.

DPD Buckets

- Current
- 1–29 DPD
- 30–59 DPD
- 60–89 DPD
- 90–119 DPD
- 120+ DPD
- Default
- Repossessed

Insights

- Majority of loans remain Current.
- Higher DPD buckets represent elevated credit risk.
- Delinquency trends support early warning monitoring.

Chapter 8: Vintage Analysis Dashboard

Vintage Month	13-24	25-36	36+	Total
2021-01			13,10,707.18	13,10,707.18
2021-02			15,64,317.58	15,64,317.58
2021-03			40,33,601.89	40,33,601.89
2021-04			26,66,968.87	26,66,968.87
2021-05			41,02,057.73	41,02,057.73
2021-06			20,51,887.52	20,51,887.52
2021-07			36,60,515.25	36,60,515.25
2021-08			37,78,608.55	37,78,608.55
2021-09			56,09,039.12	56,09,039.12
2021-10			19,72,944.58	19,72,944.58
2021-11			25,60,146.82	25,60,146.82
2021-12			28,36,189.22	28,36,189.22
2022-01			38,38,640.07	38,38,640.07
2022-02			47,32,932.24	47,32,932.24
2022-03			65,26,764.00	65,26,764.00
2022-04			31,36,182.65	31,36,182.65
2022-05			31,66,853.51	31,66,853.51
2022-06			65,19,431.39	65,19,431.39
2022-07			52,54,164.28	52,54,164.28
2022-08			54,37,175.44	54,37,175.44
2022-09			65,05,074.90	65,05,074.90
2022-10			80,73,795.09	80,73,795.09
2022-11			74,39,739.00	74,39,739.00
2022-12			1,24,08,039.60	1,24,08,039.60
2023-01			58,08,161.13	58,08,161.13
2023-02			56,56,485.20	56,56,485.20
2023-03			60,99,490.94	60,99,490.94
Total	5,12,32,754.39	13,03,15,306.69	13,62,37,141.06	31,77,85,202.14

Objective

Evaluate portfolio performance by origination cohort.

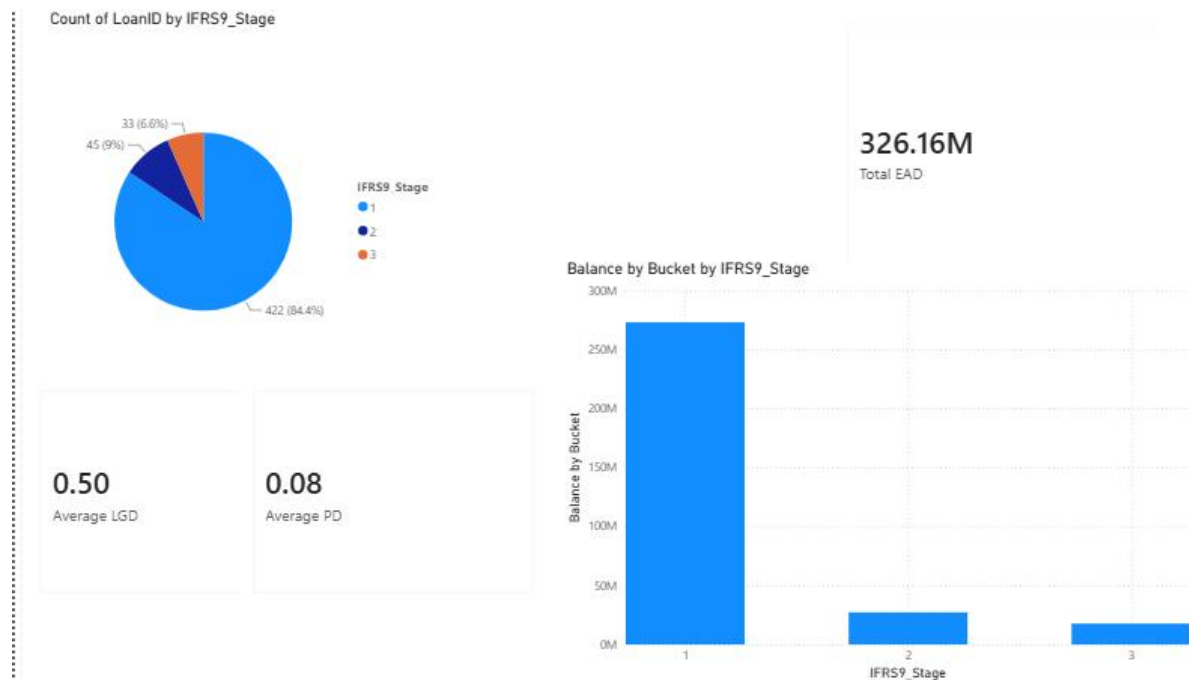
Metrics

- Months on Book
- Pool Factor
- Vintage Balance
- Net Loss Rate

Insights

- Different vintages exhibit varying performance patterns.
- Cohort analysis identifies deterioration trends.
- Vintage monitoring supports portfolio forecasting.

Chapter 9: IFRS 9 Dashboard



Objective

Support impairment reporting under IFRS 9.

Stages

Stage 1

Performing exposures.

Stage 2

Significant increase in credit risk.

Stage 3

Credit-impaired exposures.

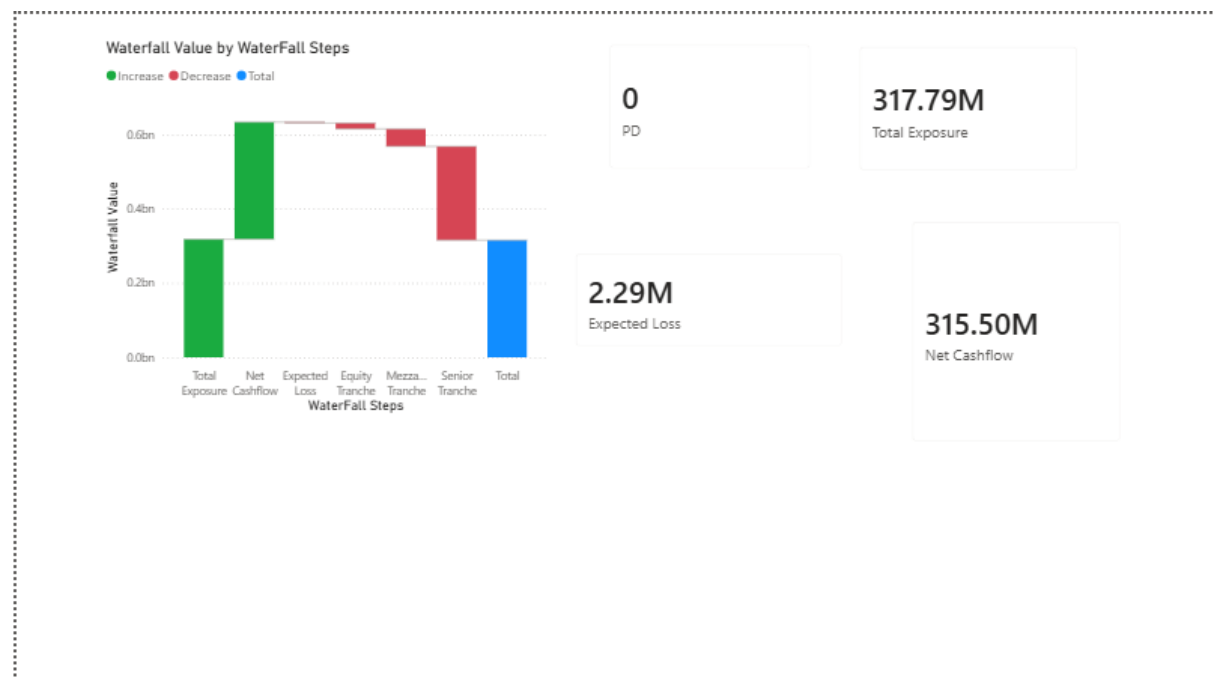
Metrics

- PD
- LGD
- EAD
- ECL

Insights

- Stage 1 dominates portfolio exposure.
 - Stage migration highlights credit deterioration.
 - ECL supports impairment provisioning.
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Chapter 10: Waterfall Dashboard



Objective

Simulate securitisation cash flow allocation.

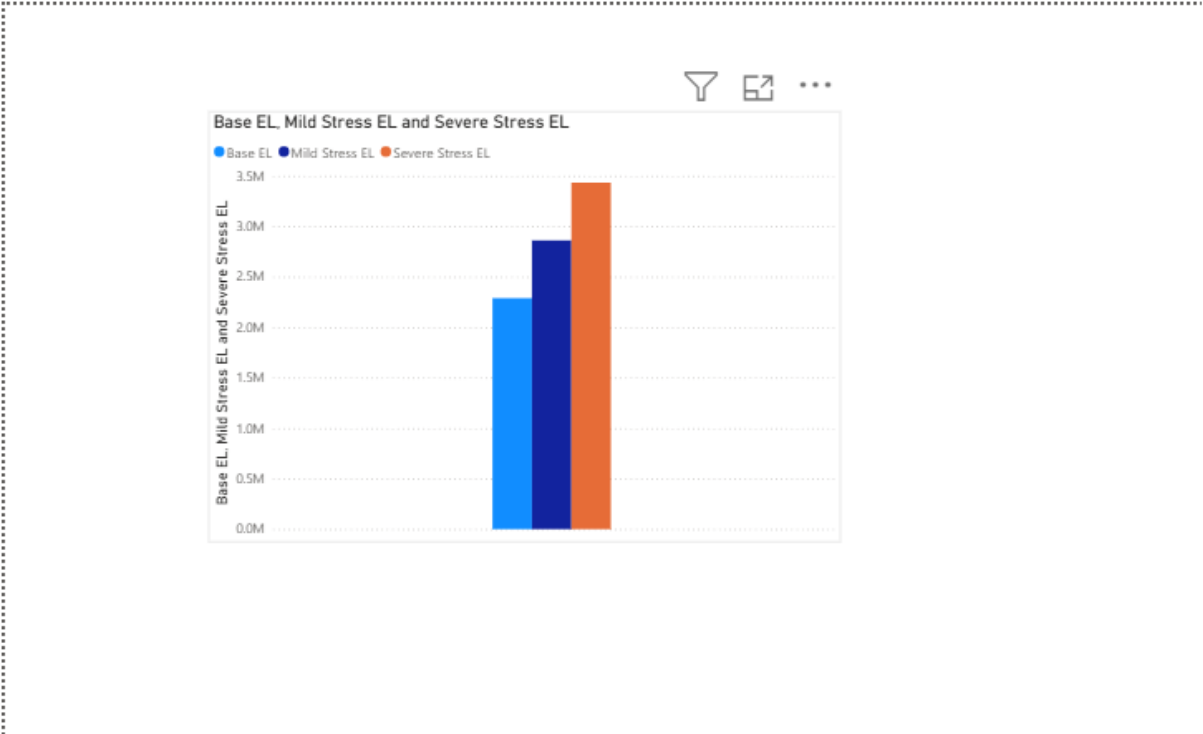
Tranches

- Senior Tranche
- Mezzanine Tranche
- Equity Tranche

Insights

- Senior investors receive payment priority.
 - Equity investors absorb first losses.
 - Waterfall structure enhances investor protection.
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Chapter 11: Stress Testing Dashboard



Objective

Evaluate portfolio resilience under stressed conditions.



Scenarios

2008 Mortgage Crisis Scenario

Assumes severe deterioration in credit quality.

Auto Loan Recession Scenario

Assumes rising delinquency and reduced collections.

Interest Rate Shock Scenario

Assumes increased borrower repayment burden.

Metrics

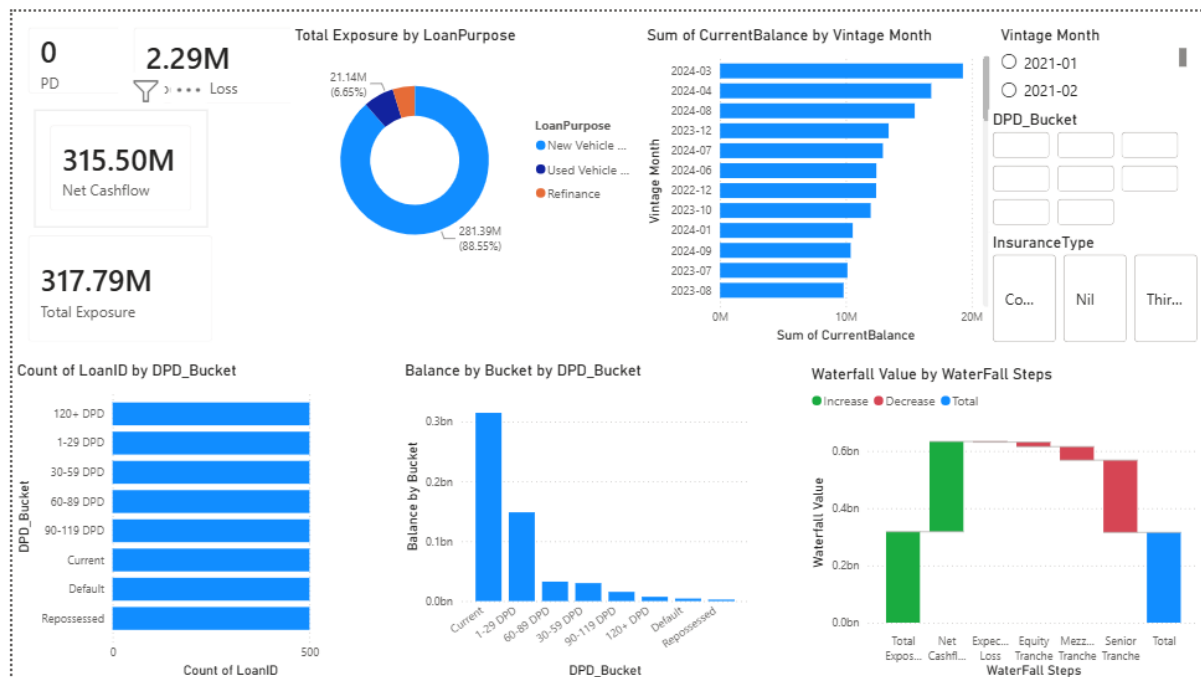
- Base Expected Loss
- Mild Stress Expected Loss
- Severe Stress Expected Loss

Insights

- Expected losses rise significantly under stress.
- Portfolio remains sensitive to macroeconomic deterioration.
- Stress testing supports risk management decisions.

Chapter 12: Key Findings

Investor Reporting



1. Portfolio exposure totals approximately 317.79M.
2. Expected Loss is estimated at 2.29M.
3. Majority of accounts remain in Current status.
4. Stage 1 exposures dominate the portfolio.
5. Stress scenarios substantially increase expected losses.

6. Vintage analysis reveals varying cohort performance.
 7. Waterfall structures effectively prioritize investor payments.
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Chapter 13: Conclusion

The Auto Loan Securitisation Analytics Dashboard successfully implements a comprehensive framework for portfolio monitoring, credit risk assessment, IFRS 9 reporting, vintage analysis, stress testing, and waterfall allocation.

The solution provides decision-makers with a unified view of securitised portfolio performance and demonstrates how Power BI can be used to support investor reporting and risk management processes in structured finance environments.

Future Enhancements

- Real-time data integration.
- Advanced roll-rate analysis.
- Machine learning-based default prediction.
- Investor performance attribution.
- Regulatory capital analytics.
- Interactive scenario simulation engine.